Exam01 Practice Questions

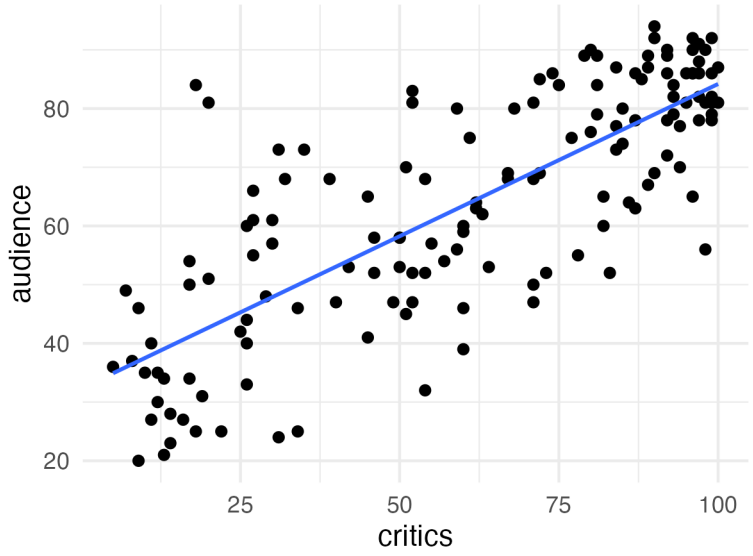
Stat 230 - Spring 2026

Practice questions for each skill on the Exam01 Review sheet.

1. Express a SLR model using proper notation

Example question In the Day 2 activity, we explored the relationship between `audience` score and `critics` score for a sample of movies. Here is the scatterplot we looked at:

```
movies <- read.csv("https://aloy.github.io/stat230-materials/data/movie_scores.csv")
gf_point(audience ~ critics, data = movies) |>
    gf_lm()
```



(a) What is the response variable? What is the explanatory variable?

(b) Write out the *theoretical* SLR model for this relationship. Define Y_i and x_i in context, include all parameters, and state the assumption placed on the error term.

Example question The Day 3 activity fit a model relating leaf width (`Width`, in mm) to `Year` for a sample of 252 leaves, giving the coefficient estimates below.

term	estimate
(Intercept)	37.723
Year	-0.018

Write out the *fitted* SLR model using proper notation.

2. Interpret the coefficients and R^2 of a model in context

```
LeafWidth <- read_csv("https://aluby.github.io/stat230/data/LeafWidth.csv")
leaf_mod <- lm(Width ~ Year, data = LeafWidth)
```

Example question Using the fitted model ($\widehat{\text{Width}}_i = 37.723 - 0.018 \times \text{Year}_i$), interpret the slope and intercept in context.

Example question For the `LeafWidth` model above, predicting `Width` from `Year` has $R^2 = 0.06$. Which of the following is the best interpretation of this value?

(a) For each additional year, leaf width increases by 0.6mm on average.
(b) About 6% of leafs have their width predicted exactly correctly by this model.
(c) About 6% of the variability in width among these leaves is explained by year.
(d) The correlation between width and year is 0.6.

3. Check necessary conditions for a regression model and discuss the implications of a model deficiency

Example question For each statement below, circle TRUE or FALSE.

(a) TRUE or FALSE: We can check the linearity condition using a scatterplot of the response against the explanatory variable, before ever fitting a model.
(b) TRUE or FALSE: A normal Q-Q plot that curves away from the diagonal line in both tails suggests the residuals are more heavily-tailed than a Normal distribution.
(c) TRUE or FALSE: A residuals-vs-fitted-values plot that fans out (wider spread for larger fitted values) suggests a violation of the equal variance condition.

(d) TRUE or FALSE: If the independence condition is violated, we can typically detect it just by looking at a residuals-vs-fitted plot.
(e) TRUE or FALSE: If the linearity condition is badly violated, the estimated slope may be misleading, even if the p-value for the slope is small.

Example question Suppose you fit a model predicting `income` from `education` for the Riverview data, and the residuals-vs-fitted plot shows a clear curved (U-shaped) pattern. Which of the following can you still trust about this model? Select all that apply, and explain your reasoning.

(a) The point estimate of the slope
(b) The p-value for the slope
(c) The standard error of the slope
(d) Predictions made from the model

4. Perform a hypothesis test to draw inference about a single regression coefficient

Example question The LeafWidth model gives the following output:

```
broom::tidy(leaf_mod) |>
    knitr::kable(digits = 3)
```

term	estimate	std.error	statistic	p.value
(Intercept)	37.723	8.575	4.399	0
Year	-0.018	0.004	-4.029	0

On $n = 252$ data points. Conduct a hypothesis test for $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$. State the test statistic, and describe (without necessarily calculating exactly) how the p-value could be found.

Example question Before collecting the NHANES sitting-time data, a researcher hypothesizes that *more* sitting time is associated with a *higher* depression risk score – not just a difference in either direction. The fitted model (`dep_score ~ sedentary_hours`) gives a slope of 0.0314 (SE = 0.0116) for `sedentary_hours`, with $df = 657$.

(a) State H_0 and H_A in symbols, matching the researcher's directional hypothesis.
(b) Is the p-value R reports by default (from `summary()`) the correct one to use here? If not, explain how you would find the correct p-value.

5. Construct and interpret confidence intervals for regression coefficients

Example question Recall Movies model: `lm(audience ~ critics)` gives a slope estimate of 0.519 (SE = 0.0345), with $df = 144$.

(a) Using the `qt()` output below, report the correct t^* value to use rounded to 3 decimal places

```
qt(.01, df = 144, lower.tail = FALSE)
```

```
[1] 2.352522
```

```
qt(.025, df = 144, lower.tail = FALSE)
```

```
[1] 1.976575
```

```
qt(.05, df = 144, lower.tail = FALSE)
```

```
[1] 1.655504
```

```
qt(.1, df = 144, lower.tail = FALSE)
```

```
[1] 1.287458
```

(b) Construct and report a 90% CI for the slope, showing the formula with numbers plugged in.

Example question Recall Riverview model using *standardized* education: `income ~ scale(education)` gives a slope estimate of 11.567 (SE = 1.613), with $df = 30$. For a 95% confidence interval, use a critical value is $t^* = 2$.

It may also be helpful know that the `income` has a mean of 53.7 and a SD of 14.5. `education` has a mean of 16 and a SD of 4.36.

(a) Construct a 95% CI for this slope.
(b) Interpret the interval in context.

6. Use a regression model to make appropriate predictions and communicate the uncertainty of these predictions through an interval

Example question Using the movies model, we predict the audience score for a movie with a critics score of 75. The point estimate is 71.2, and R produces two 90% intervals: (50.4, 92.1) and (69.3, 73.1).

(a) Which interval is the confidence interval, and which is the prediction interval? How can you tell?
(b) If we wanted to predict the audience score for *Kingsman: The Secret Service*, which has a critics score of 75, which interval should we use?
(c) For the type of interval you did *not* choose in (b), what is the correct interpretation?

Example question A researcher wants to use the NHANES `dep_score ~ sedentary_hours` model to make a claim about people who sit 6 hours per day. Explain which interval (confidence or prediction) is appropriate for each scenario below, and briefly say why.

(a) "I want a range for the depression risk score of one specific person who sits 6 hours a day."
(b) "I want a range for the *average* depression risk score among everyone who sits 6 hours a day."

7. Express an SLR model when transformations have been used on the response and/or the explanatory variables

Example question Instead of the raw PHQ-9 depression score, we actually used `dep_score`, defined as $\log(\text{PHQ-9}+1)$. The fitted model is `dep_score ~ sedentary_hours`, with intercept 0.904 and slope 0.0314.

(a) Write out the fitted SLR equation, being careful to write it in terms of `dep_score`, not the raw PHQ-9 score.
(b) A student concludes: "for every one-hour increase in sitting time, the model predicts an average increase of 0.0314 points in PHQ-9 score." Explain in one sentence why this is incorrect, then give a corrected interpretation.

Example question Suppose instead of transforming the response, we wanted to write out the general *theoretical* model for a log-transformed response, where Y_i is the raw response and x_i is the explanatory variable.

(a) Write the theoretical model using proper notation, and state the assumption on the error term.
(b) Give one reason (from what we discussed in class) why we might choose to log-transform a response variable before fitting a model.